

OptWear: AI-Driven Human-Centered Design of Wearable Health Systems

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Abstract

Wearable Internet of Things (IoT) devices can continuously monitor users' vital parameters beyond conventional clinical settings, broadening access to healthcare while reducing monitoring costs. However, adoption remains limited, partly because wearable systems are designed through *ad hoc* processes that neither account for user comfort nor efficiently explore large combinatorial hardware design spaces. Moreover, accurate evaluation through simulation or physical prototype based measurement can be expensive. We introduce OptWear, a three-stage, AI-driven framework for human-centered design of wearable health systems. The first stage efficiently explores the large combinatorial design space using multi-objective, multi-fidelity Bayesian optimization over latency and weighted energy, guided by application-level preferences. The second stage validates the predicted Pareto candidates through high-fidelity hardware measurements to identify which designs are promising for physical prototyping and human evaluation. Finally, the third stage implements those designs as wearable prototypes for user trials and usability evaluation, and learns a preference model from user feedback. This learned preference model could then be used to guide subsequent design exploration toward regions of the Pareto front that better align with user preferences. Across five health-monitoring applications and three evaluation fidelities, OptWear matches or exceeds Pareto front quality compared to state-of-the-art methods while reducing evaluation cost by 85.2%, on average. We conduct a preliminary user study to demonstrate the feasibility of collecting human feedback for preference-guided refinement.

1 Introduction

Wearable IoT devices enable continuous monitoring of key health parameters and personalized care, supporting the United Nations goal of healthy lives for all (Espay et al. 2016; Maetzler, Klucken, and Horne 2016; Daneault 2018). Despite this promise, adoption of wearable IoT devices in clinical settings has been limited (Espay et al. 2016). A key obstacle is the *ad-hoc* design process: designers select sensors, processors, and other components to meet application requirements without accounting for user and clinician preferences such as form factor, energy sustainability, and sensor

placement. These preferences are hard to incorporate because comfort and wearability cannot be reliably simulated. A device may satisfy every hardware performance metric and still be uncomfortable or difficult to use (Ozanne et al. 2018). Consequently, automated methods often ignore user preferences or rely on proxies that may not reflect them.

Closing this gap requires a human-centered design in which promising designs are evaluated by end users. Prior work achieves this by selecting components, building and instrumenting prototypes, and collecting user feedback to revise the design (Anliker et al. 2004; Rodriguez-Villegas, Iranmanesh, and Imtiaz 2018). However, repeated prototyping and user evaluation require substantial hardware, assembly, and human effort, limiting the number of explored designs. Human evaluation requires physical prototypes and participant time, limiting it to a small number of designs. This motivates using AI to narrow the design space before costly human evaluation while incorporating user preferences.

We introduce **OptWear**, a three-stage AI-driven framework for human-centered design of wearable health systems, as shown in Figure 1. The first stage formulates wearable design as a multi-objective, multi-fidelity optimization problem over latency and weighted energy. Our approach iteratively learns which designs are promising and selectively evaluates them at the appropriate fidelity to reduce the overall design evaluation cost. The search is guided by *application preferences* over the two objectives. Stage 2 physically validates the predicted Pareto candidates using high-fidelity measurements and retains the measured non-dominated designs for human evaluation. Stage 3 builds wearable prototypes of the measured Pareto designs, using 3D-printed mockups when components are unavailable, and has users wear and evaluate them. Based on their responses to the System Usability Scale (SUS) (Brooke 1996) and the short User Experience Questionnaire (UEQ-S) (Schrepp, Hinderks, and Thomaschewski 2017), OptWear learns multi-task Gaussian process (Bonilla, Chai, and Williams 2007) models of usability, pragmatic quality, and hedonic quality, and uses them to identify human-preferred regions of the Pareto front for subsequent design refinement and deployment.

We evaluate OptWear on five health-monitoring applications using an 80-component library and three evaluation fidelities: analytical models, simulation, and hardware measurements. We compare OptWear with three optimization

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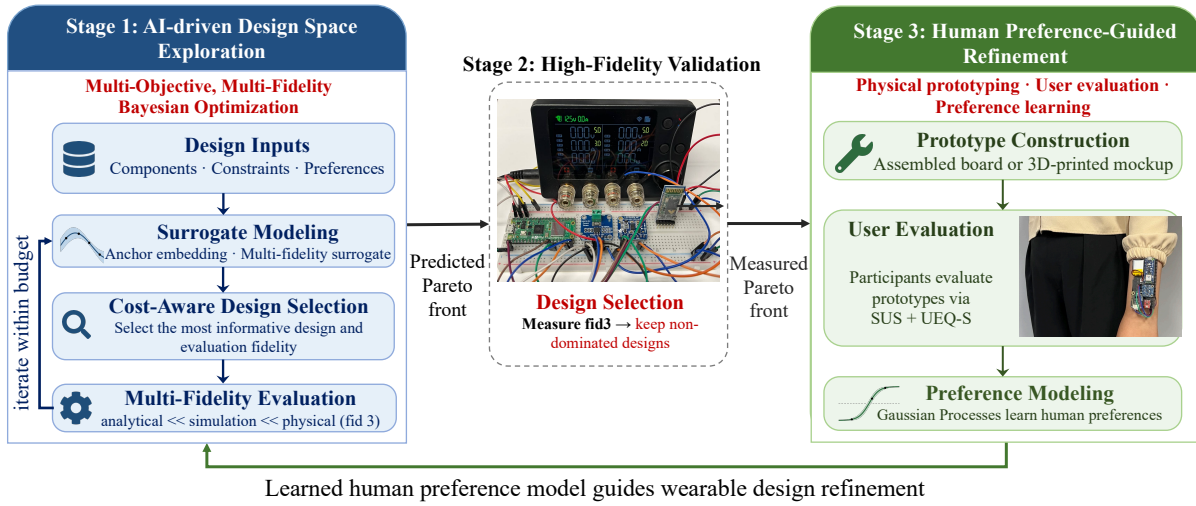


Figure 1: Overview of the three-staged OptWear framework. Stage 1 searches for promising designs using *application preferences*. Stage 2 retains measured non-dominated designs. Stage 3 learns user preferences to guide subsequent design exploration.

baselines: NSGA-II (Deb et al. 2002), MOTPE (Ozaki et al. 2020), and a Bayesian optimization (BO) baseline based on Casmopolitan (Wan et al. 2021). The evaluations show that OptWear can achieve Pareto fronts comparable to or better than baselines while using multiple fidelities to substantially reduce the overall evaluation cost with little loss in solution quality. This paper makes the following contributions:

- We introduce **OptWear**, a human-centered wearable design framework that progressively narrows large design spaces through AI-guided exploration, physical validation, and human evaluation.
- We develop a multi-objective, multi-fidelity Bayesian optimization approach that efficiently identifies promising designs while incorporating application preferences.
- We introduce a human preference-guided refinement mechanism that learns usability and user-experience preferences to guide subsequent design optimization.
- We evaluate OptWear on five health-monitoring applications with an 80-component design library and three fidelities, together with preliminary human evaluation.

2 Related Work

Prior work for wearable systems design has used evolutionary methods such as NSGA-II, AMOSA, and MOEA/D (Deb et al. 2002; Bandyopadhyay et al. 2008; Zhang and Li 2007; Anliker et al. 2004), as well as wearable-specific methods, such as DynDSE (Schiboni et al. 2020). These approaches typically require many design evaluations at a single fidelity. When comfort is considered, it is modeled by the designer rather than evaluated directly by users (Anliker et al. 2004). This motivates approaches that efficiently narrow large design spaces before incorporating direct user feedback.

BO has been applied to human-in-the-loop design (Kim et al. 2017; Liao et al. 2023; Jetton, Campbell, and Hoyle 2024; Kim, Kang, and Park 2024) and to high-dimensional

discrete spaces (Wan et al. 2021; Deshwal et al. 2023). Prior BO methods incorporate user-specified priorities over known objectives through rankings (Abdolshah et al. 2019) or importance weights (Ahmadianshalchi, Belakaria, and Doppa 2024). OptWear uses importance weights to represent application preferences during design exploration. These *application preferences* are defined over measurable objectives such as latency and weighted energy, whereas *human preferences* reflect users’ experience with the realized device and must be learned from direct evaluation. Wearable co-design methods incorporate user feedback through iterative device evaluations, but typically rely on manual design refinement rather than automated design optimization (Frances Morcillo et al. 2020; Motti and Caine 2014; Spinsante et al. 2017). To summarize, existing approaches do not integrate multi-objective, multi-fidelity design space exploration with both application preferences and human feedback, leaving a critical gap.

OptWear precisely addresses this gap through AI-guided multi-fidelity design space exploration, high-fidelity physical validation, and human preference-guided refinement of promising wearable designs.

3 Background and Problem Setup

3.1 Wearable System Design

Adoption of wearable devices in healthcare settings depends on designs that satisfy healthcare requirements, while remaining efficient and comfortable for users. Exploring these designs through multiple physical implementation is time-consuming and costly, motivating automated design space exploration. Therefore, OptWear aims to efficiently explore the design space through a multi-objective, multi-fidelity design optimization process as shown in Figure 1.

Application Representation: We model each health monitoring application as a task graph, with vertices representing requirements such as sensor type and computational capacity. The current version of OptWear includes five applications:

Motion streaming, Activity recognition, Motion and temperature monitoring, Audio event detection, and Traumatic brain injury (TBI) monitoring. Appendix B shows the task graphs.

Component Library and Objectives: OptWear constructs candidate systems from a component library $\mathcal{C}$ containing off-the-shelf components. Performance, power, size, and weight specifications are obtained from respective component datasheets. A candidate wearable design is represented by $\mathcal{X} \in \{0, 1\}^{|\mathcal{C}|}$, where $x_i = 1$ denotes selection of component i . The first stage of OptWear optimizes two objectives: application latency (L) and *weighted energy* ($E_{op}W$). The latter combines energy per operation (E_{op}) with the total system weight (W), penalizing designs that use more energy or incur higher weight while maintaining two objectives.

Preferences: OptWear uses two types of preferences with different roles. *Application preferences* specify the importance of each computed objective and are provided before the search as $\mathbf{P} = [p_{E.W}, p_L]$, where $p_{E.W} + p_L = 1$. They guide Stage 1 toward the preferred region of the Pareto front. *Human preferences* are user assessments of complete devices, including comfort, wearability, and form factor. They are collected after the device is physically built and worn in Stage 3 and then fed back into the search.

3.2 Multi-Fidelity Design Evaluation

OptWear uses three evaluation fidelities that trade off cost and accuracy.

Analytical models (fid1) estimate latency and weighted energy from datasheets and application requirements, providing the cheapest evaluations with potentially lower accuracy.

Simulation (fid2) uses the DS3 simulator (Arda et al. 2020) that schedules application tasks onto the selected components and derives the design metrics from calibrated data.

Platform measurements (fid3) evaluate a design by physically building and instrumenting it. An experienced designer needs about three hours per configuration.

OptWear jointly selects *which design* to evaluate and at *which fidelity*. Appendix C describes the measurement setup, and Appendix D reports the accuracy and relative evaluation cost of each fidelity.

3.3 Problem Formulation and Challenges

Given a component library $\mathcal{C}$ and an application, the goal in the first stage of OptWear is to identify wearable configurations that minimize weighted energy and latency:

$$\begin{aligned} \min_{\mathcal{X}} \quad & (E_{op}(\mathcal{X}) \cdot W(\mathcal{X}), L(\mathcal{X})) \\ \text{s.t.} \quad & \mathcal{X} \in \{0, 1\}^{|\mathcal{C}|}, \\ & \sum_{i=1}^{|\mathcal{C}|} x_i \leq C_{\max}, \quad \text{and } T_{op}(\mathcal{X}) \geq T_{\min}, \end{aligned} \quad (1)$$

where $C_{\max}$ limits the number of selected components and $T_{\min}$ specifies the minimum required battery life.

This OptWear problem presents four key challenges. First, binary component selection creates a high-dimensional combinatorial search space, making exhaustive search impractical and ruling out direct gradient-based optimization. Second, the high-dimensional discrete representation makes it

challenging to construct surrogate models that generalize effectively across candidate designs. Third, relying solely on high-fidelity platform evaluations is costly, while relying on lower fidelities can lead to inaccurate estimates of true high-fidelity performance. Fourth, none of the three fidelities can evaluate whether a device is comfortable to wear, which can only be determined after it has been built and tested.

Real-World Design Complexity: The design spaces for the health-monitoring applications in our evaluation range from 2,800 to 4.7 million configurations (Appendix E). Assuming approximately three hours for an experienced designer to implement and evaluate one configuration at *fid3*, exhaustive evaluation would require hundreds to more than 500k days.

4 OptWear: Human-Centered Wearable Design Optimization

This section describes the details of OptWear, a three-stage AI-driven framework for resource-efficient wearable system design that incorporates both application-level preferences and human preferences, as shown in Figure 1.

4.1 AI-driven Design Space Exploration

Stage 1 in OptWear is an iterative BO approach. In each iteration, we jointly select the wearable configuration and evaluation fidelity under a limited resource budget while incorporating application-level hardware objective preferences.

Design representation and surrogate modeling. Wearable configurations form a discrete combinatorial design space. OptWear maps each discrete configuration to a continuous representation based on its distances to a set of structurally diverse anchor designs, enabling Gaussian process (GP)-based surrogate modeling. A multi-fidelity GP (Le Gratiet 2013) is learned for each objective by jointly modeling observations collected across evaluation fidelities. The design representation and surrogate construction are detailed in Appendix A.1.

Information-theoretic design and fidelity selection. At each BO iteration, OptWear selects the design $\mathcal{X}$ and fidelity f that provides the most information about the unknown highest-fidelity Pareto front $\mathcal{F}^*$ relative to evaluation cost:

$$\beta(\mathcal{X}, f) = \frac{I(\mathcal{O}^f(\mathcal{X}); \mathcal{F}^* | \mathcal{D})}{\lambda^f}, \quad (2)$$

where $\mathcal{D}$ denotes the evaluations collected so far, $\mathcal{O}^f(\mathcal{X})$ is the vector of objective values at fidelity f , and λ^f is the normalized evaluation cost. Inspired by information-theoretic multi-objective acquisition function from (Belakaria, Deshwal, and Doppa 2019), the information gain is approximated using posterior samples of the highest-fidelity Pareto front.

Incorporating application preferences. Each application may prioritize the hardware objectives differently. OptWear represents these known priorities using application-level preference weights $\mathbf{p} = \{p_1, \dots, p_N\}$, where $p_j \geq 0$ and $\sum_j p_j = 1$. These weights control the relative contribution of each objective to the information-gain acquisition, directing the evaluation budget toward trade-offs that are more relevant to the target application. Theoretical details of the acquisition approximation, preference weighting, and optimization procedure are provided in Appendix A.2.

4.2 High-Fidelity Validation

The Pareto-front identified at the end of Stage-1 includes designs that have estimations of the hardware objective values. These objectives may present a gap when compared to their real-world implementation due to estimation error in the surrogate models. Consequently, some of the designs may not be in the Pareto front when evaluated on the real hardware. Full-scale prototyping and user studies incurs high cost without the benefits of Pareto optimality. Therefore, Stage-2 physically validates the predicted Pareto candidates from Stage 1 using high-fidelity measurements at *fid3*, reusing any high-fidelity results already available. The Pareto front is then recomputed from the measured objectives, and the resulting non-dominated designs are passed to Stage-3.

4.3 Human Preference-Guided Refinement

Hardware objectives alone do not fully capture subjective factors that influence the usability and acceptance of wearable systems. Therefore, Stage 3 models human evaluations of the physically validated Pareto designs and uses the learned human preference models to guide design-refinement.

Human-factor modeling. We consider three human-centered metrics: SUS usability, UEQ-S pragmatic quality, and UEQ-S hedonic quality. Let $h_{u,k}^{(q)}$ denote the score provided by user u for design k on human factor q , and define $\mathbf{Z}_k = [\text{CONT}(\mathcal{X}_k)^\top, \mathbf{Y}_k^\top]^\top$, where $\mathbf{Y}_k$ is the high-fidelity hardware objectives vector. For each human factor q , OptWear learns a multi-task GP, with users treated as related tasks (Bonilla, Chai, and Williams 2007):

$$h_{u,k}^{(q)} = g_q(\mathbf{Z}_k, u) + \epsilon_{u,k}^{(q)}, \quad \epsilon_{u,k}^{(q)} \sim \mathcal{N}(0, \sigma_q^2). \quad (3)$$

The multi-task formulation captures user-specific evaluations while allowing information to be shared across related users.

For each posterior Pareto sample, the learned human-factor models estimate the relative evaluation of competing designs for each user and human factor. These probabilities are aggregated across human factors and users, and the highest-ranked designs whose cumulative normalized score reaches ρ form the human-guided Pareto sample $\tilde{\mathcal{F}}_H^{*(t)}$; details are provided in Appendix A.3.

Human-guided information acquisition. The human-guided Pareto samples become the information target for the next optimization round:

$$\beta_{\text{HIL}}(\mathcal{X}, f) = \frac{I(\mathcal{O}^f(\mathcal{X}); \mathcal{F}_H^* \mid \mathcal{D}, \mathcal{D}_H)}{\lambda^f}, \quad (4)$$

where $\mathcal{F}_H^*$ denotes the human-guided portion of the unknown highest-fidelity Pareto front and $\mathcal{D}_H$ denotes the human-evaluation data. Therefore, human feedback directs subsequent design optimization procedure toward reducing uncertainty in the human-guided region of the Pareto front.

5 Experiments and Results

5.1 Experimental Setup

Applications and component library: We evaluate OptWear on the five health-monitoring applications intro-

duced in Section 3, which use common wearable health-processing kernels. The component library includes 80 components: 35 processors, 19 accelerometers, 16 communication modules, 5 temperature sensors, and 5 microphones.

We compare OptWear with NSGA-II (Deb et al. 2002) and MOTPE (Ozaki et al. 2020), which represent evolutionary and model-based multi-objective optimization methods, and with a state-of-the-art multi-objective BO approach (Belakaria, Deshwal, and Doppa 2019; Wan et al. 2021) that employs Casmopolitan surrogate to model discrete design variables under a single-fidelity acquisition function. None of these baselines selects among fidelities, so we compare against them at *fid2*, which is accurate enough to be meaningful while cheap enough to run a full single-fidelity search for every method. OptWear uses 15 anchor designs, which gives the best average result among the values we tested (Appendix G). NSGA-II uses PyMOO (Blank and Deb 2020), MOTPE uses Optuna (Akiba et al. 2019), and all BO methods are implemented in PyTorch (Paszke et al. 2019).

Validation protocol and metrics: Each method is run with four random seeds per application under a budget of 50 evaluations. The primary metric is the Pareto hypervolume (PHV) of the final front, with the reference point set by the maximum weighted energy and latency observed across all methods and seeds, scaled by 1.1. PHVs are comparable across methods within an application but not across applications. The secondary metric is resource cost, obtained by multiplying the number of evaluations at each fidelity by that fidelity’s relative cost. Because multi-fidelity runs evaluate designs at different fidelities, we re-evaluate the final Pareto front at *fid2* before computing PHV, which places all methods on a common scale and removes any effect of surrogate prediction error. We note that the re-evaluation is used only for benchmarking. We report the mean and standard deviation across seeds in the following sections.

5.2 Optimization Performance

We first compare OptWear with the baselines using a single evaluation fidelity (*fid2*). Table 1 shows the mean PHV and standard deviation for each method and application.

OptWear obtains the highest mean hypervolume on four of the five applications. Compared with NSGA-II, the improvement ranges from 0.47% on audio event detection to 15.63% on activity recognition. Compared with Casmopolitan, the difference ranges from -0.47% on audio event detection to 11.04% on motion and temperature monitoring. Casmopolitan is best on audio event detection, with the smallest design space at 2,800 configurations, where all methods agree within 1%. The differences from MOTPE are smaller, ranging from -0.23% to 0.74%, with three applications showing differences below 0.1%. Since these differences fall within the seed-to-seed variation, OptWear and MOTPE reach comparable front quality at a single fidelity. However, unlike the baselines, OptWear can use multiple fidelities and minimize resource cost, as we show next.

The standard deviations separate the methods more clearly. OptWear stays below 5.31 on every application, while Casmopolitan varies by 142.78 on motion and temperature monitoring and NSGA-II by 10.79 on TBI monitoring.

Method	Motion streaming	Activity recognition	Motion & temperature	Audio event detection	Traumatic brain injury
Casmopolitan	25.68±2.67	26.26±0.92	2005.70±142.78	4.30 ±0.006	423.33±3.38
MOTPE	27.20±0.03	27.03±0.06	2225.87±1.87	4.29±0.009	424.64±0.27
NSGA-II	23.89±2.25	23.55±5.17	1974.27±11.65	4.25±0.029	405.33±10.79
OptWear	27.22 ±0.01	27.23 ±0.09	2227.13 ±5.3	4.28±0.027	424.88 ±0.14

Table 1: Pareto hypervolume (s-J-g) obtained with a single fidelity (*fid2*), mean $\pm$ std over four seeds.

Application	Cost saved (%)	PHV loss (%)
Motion streaming	77.4±6.49	0.01±0.05
Activity recognition	84.0±1.04	0.02±0.31
Motion and temp.	87.0±5.20	−0.22±0.94
Audio event detection	88.2±3.12	1.06±1.24
Traumatic brain injury	89.4±1.04	1.59±1.09
Average	85.2±3.38	0.49±0.73

Table 2: Resource cost savings and PHV loss of multi-fidelity search (*fid1* and *fid2*) relative to single-fidelity *fid2* search (mean $\pm$ std over four seeds).

5.3 Multi-Fidelity Optimization

The comparison in Table 1 uses a single fidelity, which is the only setting supported by the baselines. We now allow OptWear to choose between *fid1* and *fid2* at each iteration and compare it against its single-fidelity *fid2* using the same budget and the same four seeds. We report the percentage of resource cost saved and the percentage of hypervolume lost relative to that run. Table 2 shows that multi-fidelity search reduces resource cost by 77.4% to 89.4%, averaging 85.2%, while PHV changes by less than 1.6% in every case and by 0.49% on average. On motion and temperature monitoring, the PHV loss is −0.22%, which means that the multi-fidelity front is slightly better than the single-fidelity front. The acquisition function achieves this by spending majority of its budget on *fid1*, which costs a tenth of *fid2*, and reserving *fid2* for the few candidates whose evaluation is most informative about the Pareto front.

Next, we utilize all three fidelities, including physical measurement at *fid3*. A single-fidelity *fid3* baseline is not feasible because each evaluation requires a prototype and only a subset of library components is available, leaving a small number of designs for a meaningful search. We compare the measured cost of the three-fidelity search against 50 *fid3* evaluations with a cost of 5000. A design selected for *fid3* is measured only when all of its components are available; otherwise, we skip it and evaluate the next highest-ranked design–fidelity pair. Across the five applications, the three-fidelity search costs 140 to 383, with an average of 247. This corresponds to savings of 92.3% to 97.2%, or 95.1% on average, when compared to the projected cost of a single *fid3* search. The optimizer therefore uses *fid3* only for the few designs where physical measurement is most informative about the Pareto front. Appendix J reports the per-application costs. Additional results on scalability of the multi-fidelity search and resource cost savings are also reported in Appendix I.

5.4 Preference-Aware Optimization

Application preferences let a designer direct the search toward parts of the Pareto front that satisfy the preferences. We evaluate five preference pairs for $(p_{E.W}, p_L)$, from (0, 1), which optimizes latency alone, to (1, 0), which optimizes weighted energy alone. For each pair, we score every design on the resulting front by the weighted sum of its normalized objectives and report the lowest-scoring design, which we refer to as the rank-1 design, as defined further in Appendix I.

Table 3 shows the rank-1 designs for across all applications and preference pairs. Moving from the latency-only preference to the weighted-energy-only preference, the weighted energy of the selected design decreases on all five applications while its latency rises, which shows that the preference vector moves the selected design along the front in the desired direction. The intermediate preferences do not always follow a strict order because the Pareto front is discrete, which means small changes in the weights may not lead to selection of a different design. TBI monitoring application shows the largest deviation, with the lowest weighted energy occurring at preference 4 rather than at the weighted-energy-only preference. This is because the two objectives are not strictly opposed: a design with low latency does not necessarily have high weighted energy. As a result, a design selected while favoring one objective can also perform well on the other. Appendix H reports show the same effect across the whole front and the distribution of both objectives.

6 Path to Deployment

This section describes our plan to deploy OptWear designs in healthcare settings. We start by incorporating user preferences into the design process and performing a pilot deployment for user feedback.

6.1 Incorporating User Preferences

Deployment of OptWear adds human preferences to the optimization process. Recall that the first stage optimizes latency and weighted energy. Neither captures whether a device is comfortable to wear, which can only be determined by users after the device is built. The third stage therefore builds the designs from the Pareto front, asks users to evaluate them with standardized questionnaires, and uses the responses to learn a preference model that is fed back into the search.

We conducted an initial user evaluation with six participants to establish the questionnaire and data-collection procedure. Each participant tried two implemented designs and completed a single set of questionnaires covering both designs. Because our current IRB protocol permits only one

Preference		Applications				
$p_{E \cdot W}$	p_L	Motion streaming	Activity recognition	Motion & temperature	Audio event detection	Traumatic brain injury
0.00	1.00	2.60 / 328	99.49 / 450	1595.75 / 1194	2.79 / 66	13.45 / 5477
0.25	0.75	1.35 / 321	8.45 / 750	1227.03 / 1029	11.98 / 107	1359.30 / 1524
0.50	0.50	1.17 / 366	4.88 / 768	158.40 / 1021	0.18 / 673	42.64 / 1649
0.75	0.25	0.02 / 342	4.90 / 1506	1.55 / 1127	0.46 / 845	1.24 / 5568
1.00	0.00	0.03 / 5142	0.41 / 1903	50.92 / 9539	0.46 / 845	4.10 / 6065

Table 3: Rank-1 design under application preference pairs, evaluated at *fid2*, reported as weighted energy (mJ·g) / latency (ms).

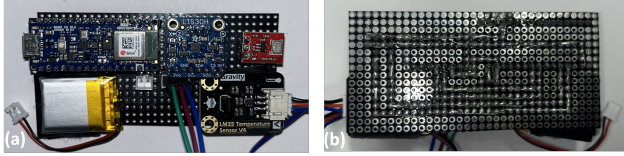


Figure 2: A design for the traumatic brain injury monitoring application, shown from (a) the top and (b) the back side.

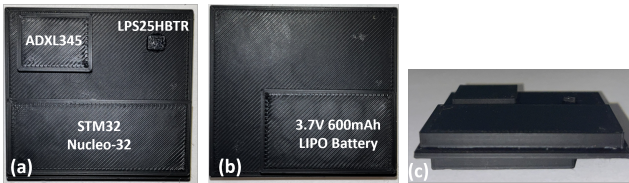


Figure 3: A 3D-printed mockup of a motion and temperature monitoring design from the (a) top, (b) bottom, and (c) side.

combined instrument rather than a separate survey for each design, the responses cannot be attributed to individual designs. A revised protocol that allows separate surveys for each design is currently under review.

Instruments. We use two standardized questionnaires. The System Usability Scale (SUS) (Brooke 1996) contains ten statements rated from strongly disagree to strongly agree. It produces an overall score from 0 to 100, as well as usability and learnability subscales. The short User Experience Questionnaire (UEQ-S) (Schrepp, Hinderks, and Thomaschewski 2017) contains eight pairs of opposing adjectives rated on a seven-point scale. It produces a pragmatic quality score for task-related aspects, a hedonic quality score for appeal and novelty, and an overall mean score. UEQ-S scores range from -3 to $+3$.

Implementing designs for evaluation. Designs shown to users cannot use *fid3* because its bench supply and INA219 tether the device, while breadboard assemblies are not wearable. We therefore build a separate battery-powered version on a single board, as shown in Figure 2. If the required components are not physically available, we instead 3D print a mockup of the design, as shown in Figure 3. The mockup matches the design’s size, mass, and component placement. This allows us to collect responses for Pareto-front designs with unavailable components.

Results. Six participants tried the implemented designs, one

of which is shown in Figure 2, and completed both questionnaires. The mean SUS score is 61.7 ± 20.6 , slightly below the typical value, while the mean UEQ-S score is 1.44 ± 0.87 , which is in the positive range. Hedonic quality for this design is higher than pragmatic quality (1.75 ± 0.89 against 1.12 ± 1.40), indicating more positive responses to the design’s appeal than to its usability. Responses vary considerably, with SUS scores ranging from 22.5 to 75.0. Appendix K reports the individual scores.

The next step is to collect per-design responses from multiple users across several physically validated Pareto designs and learn human preference models. For each of the three human factors: SUS usability, UEQ-S pragmatic quality, and UEQ-S hedonic quality. We fit a multi-task GP using users as related tasks and the design representation together with the high-fidelity hardware objectives as inputs. The learned models are then used to rank posterior Pareto samples across human factors and users, identify human-guided regions of the Pareto front, and guide the next optimization round toward designs that better align with user preferences. The human preference-aware OptWear will be included in our full-scale deployment with patients in healthcare settings at Providence hospital in Spokane, WA. Alongside the study, we are expanding the component inventory, which will enable a single-fidelity *fid3* baseline (Appendix L).

7 Conclusion and Future Work

This paper presented OptWear, a three-stage framework for designing wearable health systems. Stage 1 searches the combinatorial space of component selections with multi-objective, multi-fidelity BO. Stage 2 physically validates the predicted Pareto candidates using *fid3* measurements and retains the measured non-dominated designs for human evaluation. Finally, Stage 3 evaluates the implemented wearable prototypes with users, learns human preference models from their feedback, and uses the learned preferences to guide subsequent design optimization toward human-guided regions of the Pareto front. Across five applications and an 80-component library, Stage 1 of OptWear outperforms all baseline approaches in Pareto front quality on four out of five applications, and multi-fidelity search substantially reduces evaluation cost. We also deploy two designs with users and obtain usability scores. Our immediate next steps are to expand the component inventory and to collect per-design survey responses for full-scale deployment in health settings.

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A Additional Details of OptWear

This appendix provides additional technical details for the design representation, information-theoretic acquisition, and human preference-guided refinement described in Section 4.

A.1 Continuous Representation and Multi-Fidelity Surrogates

A wearable configuration is represented by a binary component-selection vector $\mathcal{X} \in \{0, 1\}^{|\mathcal{C}|}$, where $\mathcal{C}$ denotes the component library. To enable GP modeling over the combinatorial design space, OptWear maps each configuration to its Hamming distances from a set of m structurally diverse anchor designs $\mathcal{A} = \{\mathbf{a}_1, \dots, \mathbf{a}_m\}$:

$$\text{CONT}(\mathcal{X}) = (d_H(\mathcal{X}, \mathbf{a}_1), \dots, d_H(\mathcal{X}, \mathbf{a}_m))^T \in \mathbb{R}^m, \quad (5)$$

where

$$d_H(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^{|\mathcal{C}|} [u_i \neq v_i]. \quad (6)$$

The anchors span different sparsity levels of the design space so that the embedding preserves variation in component-selection patterns. The resulting continuous representation permits the use of standard GP kernels with automatic relevance determination.

For each objective o_j , OptWear trains a separate multi-fidelity GP over $\text{CONT}(\mathcal{X})$ and the evaluation fidelity. The surrogate jointly models observations obtained at all available fidelities and provides the predictive mean and uncertainty required by the acquisition function.

A.2 Application-Guided Information-Theoretic Acquisition

At each BO iteration, the objective is to select the design-fidelity pair that provides the largest expected information gain about the unknown highest-fidelity Pareto front per unit cost:

$$\beta(\mathcal{X}, f) = \frac{I(\mathcal{O}^f(\mathcal{X}); \mathcal{F}^* \mid \mathcal{D})}{\lambda^f}. \quad (7)$$

Inspired by information-theoretic multi-objective BO (Belakaria, Deshwal, and Doppa 2019), we approximate the distribution of the unknown Pareto front using T posterior Pareto samples $\tilde{\mathcal{F}}^{*(1)}, \dots, \tilde{\mathcal{F}}^{*(T)}$. The resulting acquisition is approximated as

$$\beta(\mathcal{X}, f) \approx \frac{1}{\lambda^f} \sum_{j=1}^N \left[\log \sigma_j^f(\mathcal{X}) - \frac{1}{T} \sum_{t=1}^T \log \sigma_{j,t}^f(\mathcal{X}) \right], \quad (8)$$

where $\sigma_j^f(\mathcal{X})$ is the predictive uncertainty for objective j and $\sigma_{j,t}^f(\mathcal{X})$ is the corresponding uncertainty conditioned on the t -th posterior Pareto sample.

To incorporate known application-level application priorities, let $\mathbf{p} = \{p_1, \dots, p_N\}$ satisfy $p_j \geq 0$ and $\sum_{j=1}^N p_j = 1$. The preference-aware acquisition becomes

$$\beta_{\mathbf{p}}(\mathcal{X}, f) \approx \frac{1}{\lambda^f} \sum_{j=1}^N p_j \left[\log \sigma_j^f(\mathcal{X}) - \frac{1}{T} \sum_{t=1}^T \log \sigma_{j,t}^f(\mathcal{X}) \right]. \quad (9)$$

Objectives with larger p_j therefore contribute more strongly to the selection criterion. OptWear maximizes $\beta_{\mathbf{p}}(\mathcal{X}, f)$ using multi-start local search with 15 restarts and evaluates the highest-scoring design-fidelity pair. The new observation is added to $\mathcal{D}$ and the multi-fidelity GP surrogates are updated. The process continues until the evaluation budget for the current optimization round is exhausted, after which the estimated highest-fidelity Pareto front is passed to Stage 2.

A.3 Human-Factor Modeling and Aggregation

Let

$$\mathcal{P}_H = \{(\mathcal{X}_k, \mathbf{Y}_k)\}_{k=1}^K \quad (10)$$

denote the Pareto designs validated at *fid3* in Stage 2, where $\mathbf{Y}_k$ is the corresponding high-fidelity technical outcome vector. Let $\mathcal{K}_u$ denote the subset of these designs evaluated by user u . We consider $Q = 3$ human factors: SUS usability, UEQ-S pragmatic quality, and UEQ-S hedonic quality. The score provided by user u for design k on factor q is denoted by $h_{u,k}^{(q)}$.

Augmented human-factor input. The subjective evaluation of a wearable system can depend on both its configuration and its technical performance. We therefore augment the continuous design representation with the corresponding high-fidelity technical outcomes:

$$\mathbf{Z}_k = [\text{CONT}(\mathcal{X}_k)^T, \mathbf{Y}_k^T]^T. \quad (11)$$

All continuous dimensions of $\mathbf{Z}$ are standardized before GP fitting to account for differences in scale between the design embedding and technical outcomes; the same transformation is used for prediction.

For each human factor q , the available human-evaluation data are

$$\mathcal{D}_H^{(q)} = \{((\mathbf{Z}_k, u), h_{u,k}^{(q)}) : u = 1, \dots, U, k \in \mathcal{K}_u\}. \quad (12)$$

The complete human-evaluation dataset is $\mathcal{D}_H = \{\mathcal{D}_H^{(q)}\}_{q=1}^Q$.

Multi-task human-factor models. For each human factor q , OptWear learns a separate multi-task GP, with users treated as related tasks (Bonilla, Chai, and Williams 2007):

$$h_{u,k}^{(q)} = g_q(\mathbf{Z}_k, u) + \epsilon_{u,k}^{(q)}, \quad \epsilon_{u,k}^{(q)} \sim \mathcal{N}(0, \sigma_q^2). \quad (13)$$

Following the multi-task covariance formulation, the covariance for factor q is

$$K^{(q)}((\mathbf{Z}, u), (\mathbf{Z}', u')) = k_{\mathbf{Z}}^{(q)}(\mathbf{Z}, \mathbf{Z}') B_{uu'}^{(q)}, \quad (14)$$

where $k_{\mathbf{Z}}^{(q)}(\cdot, \cdot)$ is a covariance function over the augmented design input and $\mathbf{B}^{(q)}$ is a learned positive-semidefinite task covariance matrix across users. The input kernel therefore models similarity between wearable configurations and their technical outcomes, while $\mathbf{B}^{(q)}$ captures dependencies among users for human factor q .

For factor q and user u , the posterior provides mean $\mu_{q,u}(\mathbf{Z})$, variance $\sigma_{q,u}^2(\mathbf{Z})$, and covariance $c_{q,u}(\mathbf{Z}, \mathbf{Z}')$.

Human-factor comparisons over posterior Pareto samples. Because the human-factor models depend on both the design and its technical outcomes, OptWear retains the correspondence between designs and outcomes within each posterior Pareto sample. Let

$$\tilde{\mathcal{P}}^{*(t)} = \left\{ \left(\mathcal{X}_i^{(t)}, \mathbf{Y}_i^{(t)} \right) \right\}_{i=1}^{M_t} \quad (15)$$

denote the t -th posterior Pareto sample, where M_t is the number of designs in that sample. Here $\mathbf{Y}_i^{(t)}$ is the sampled highest-fidelity technical outcome associated with design $\mathcal{X}_i^{(t)}$.

For each sampled candidate, define

$$\mathbf{Z}_i^{(t)} = \left[\text{CONT}(\mathcal{X}_i^{(t)})^\top, (\mathbf{Y}_i^{(t)})^\top \right]^\top. \quad (16)$$

For two candidates i and j , $g_q(\mathbf{Z}_i^{(t)}, u) - g_q(\mathbf{Z}_j^{(t)}, u)$ is Gaussian under the GP posterior. For compactness, define

$$\begin{aligned} \mu_{q,u,i}^{(t)} &= \mu_{q,u}(\mathbf{Z}_i^{(t)}), \\ \sigma_{q,u,i}^{2,(t)} &= \sigma_{q,u}^2(\mathbf{Z}_i^{(t)}), \\ c_{q,u,ij}^{(t)} &= c_{q,u}(\mathbf{Z}_i^{(t)}, \mathbf{Z}_j^{(t)}). \end{aligned} \quad (17)$$

The probability that user u evaluates candidate i more favorably than candidate j with respect to human factor q is

$$\begin{aligned} P_{q,u}^{(t)}(i \succ j) &= P\left(g_q(\mathbf{Z}_i^{(t)}, u) > g_q(\mathbf{Z}_j^{(t)}, u) \mid \mathcal{D}_H^{(q)}\right) \\ &= \Phi\left(\frac{\mu_{q,u,i}^{(t)} - \mu_{q,u,j}^{(t)}}{\sqrt{\sigma_{q,u,i}^{2,(t)} + \sigma_{q,u,j}^{2,(t)} - 2c_{q,u,ij}^{(t)}}}\right). \end{aligned} \quad (18)$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. The selected human factors are oriented such that larger scores correspond to more favorable evaluations.

Aggregation across human factors and users. For each candidate i , OptWear first computes its human-factor ranking score for user u by aggregating the posterior comparison probabilities across the Q human factors and all competing designs in the sampled Pareto front:

$$R_u^{(t)}(i) = \frac{1}{Q} \sum_{q=1}^Q \frac{1}{M_t - 1} \sum_{\substack{j=1 \\ j \neq i}}^{M_t} P_{q,u}^{(t)}(i \succ j). \quad (19)$$

We use equal weighting across the human factors in the absence of prior information specifying different relative importance.

The group-level human-factor ranking score is then obtained by aggregating the user-specific ranking scores:

$$R^{(t)}(i) = \frac{1}{U} \sum_{u=1}^U R_u^{(t)}(i). \quad (20)$$

Thus, individual user responses are modeled before group-level aggregation; the aggregation operates on user-specific posterior comparison evidence.

The ranking scores are normalized within each posterior Pareto sample:

$$S_H^{(t)}(i) = \frac{R^{(t)}(i)}{\sum_{j=1}^{M_t} R^{(t)}(j)}. \quad (21)$$

Let $i_{(1)}, \dots, i_{(M_t)}$ denote the candidate indices sorted in descending order of $S_H^{(t)}$. OptWear retains the smallest highest-ranked subset whose cumulative normalized human-factor ranking score reaches ρ :

$$k_t = \min \left\{ k : \sum_{\ell=1}^k S_H^{(t)}(i_{(\ell)}) \geq \rho \right\}. \quad (22)$$

The resulting human-guided posterior Pareto samples are used as the information target in Equation 4 during the subsequent optimization round. Newly identified Pareto designs are subsequently validated at *fid3* in Stage 2 before additional human evaluations are collected and the human-factor models are updated.

B Application Task Graphs

Figure 4 shows the task graphs of the five health-monitoring applications used in our evaluation. Vertices correspond to sensing, processing, and communication requirements that the generated wearable design must support.

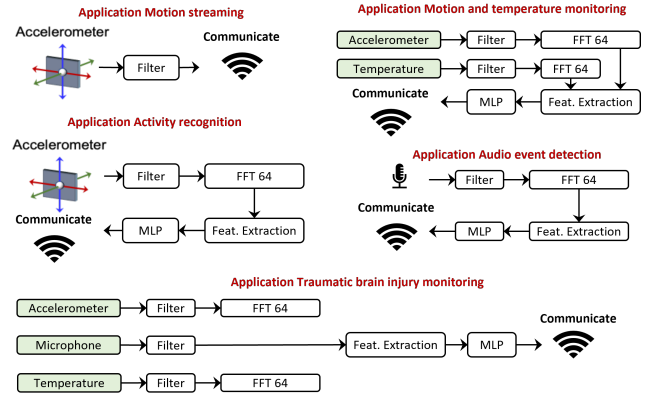


Figure 4: Task graphs of the five health-monitoring applications.

C Platform Measurement Setup

Platform measurements (*fid3*) require a physical implementation of the design. The processor options include embedded platforms and microcontroller boards such as Raspberry Pi (Raspberry Pi Ltd 2023), Odroid-XU3 (Hardkernel Co., Ltd. 2014), Nvidia Jetson (NVIDIA Corporation 2020), and Arduino (Arduino LLC 2015). To implement a design, we acquire the components, check their datasheets for electrical compatibility, connect them on a prototyping board, and implement the application in C or Python. We then measure latency on the device and power using an INA219 (Texas Instruments 2015) current sensor. Figure 5 shows an example design and its measurement setup.

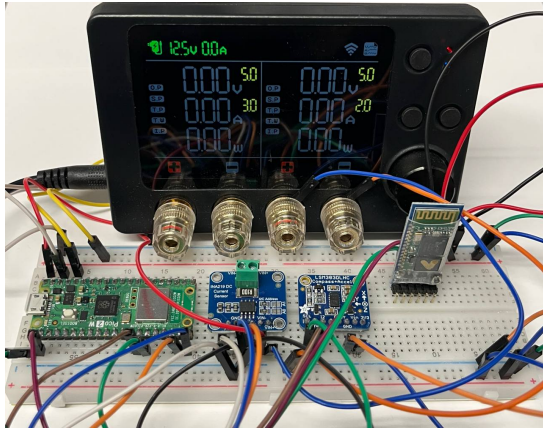


Figure 5: A design for the motion streaming application, assembled and instrumented for measurement at *fid3*.

D Fidelity Accuracy Characterization

We calculate the accuracy of each fidelity with respect to the highest fidelity (*fid3*). For each application we implement five designs on hardware and measure their latency and weighted energy at *fid3*. For each design i of application j , the relative errors of a lower-fidelity model $F_x \in \{fid1, fid2\}$ are given by Equations 23 and 24:

$$e_{L,(i,j)} = \left| \frac{L_{fid3,(i,j)} - L_{F_x,(i,j)}}{L_{fid3,(i,j)}} \right|, \quad (23)$$

$$e_{E \cdot W,(i,j)} = \left| \frac{(E \cdot W)_{fid3,(i,j)} - (E \cdot W)_{F_x,(i,j)}}{(E \cdot W)_{fid3,(i,j)}} \right|. \quad (24)$$

We aggregate these errors into a mean absolute percentage error (MAPE) for each objective separately, and combine them into the single accuracy score of Equation 25:

$$A_{total} = \sqrt{(1 - MAPE_L)(1 - MAPE_{E \cdot W})}. \quad (25)$$

Using Equation 25, *fid1* reaches an accuracy of 0.4 and *fid2* reaches 0.9 relative to *fid3*. The relative evaluation costs of *fid1*, *fid2*, and *fid3* are 1, 10, and 100.

E Design Space Sizes

Table 4 lists the number of valid configurations for each application with our 80-component library, together with the time an exhaustive search at the highest fidelity would require. The number of component types an application needs determines the size of its design space, so traumatic brain injury monitoring, which combines three sensing branches, has by far the largest space.

F Scalability of the Embedding

The anchor-distance embedding maps each design to a fixed 15-dimensional space, so the surrogate always operates in 15 dimensions regardless of the size of the component library. Without the embedding, the Gaussian process uses the raw binary design vector, so its input dimension grows with the

Application	Configurations	Est. time (days)
Motion streaming	10,640	1,330
Activity recognition	10,640	1,330
Motion and temperature	957,600	119,700
Audio event detection	2,800	350
Traumatic brain injury	4,788,000	598,500

Table 4: Design space size and estimated time to implement all configurations, at three hours per configuration.

number of components. To measure the effect, we create synthetic libraries with 160, 240, 320, and 400 components by replicating the original 80 components and scaling their values by random factors drawn from a Gaussian distribution. We then run both variants on each library size.

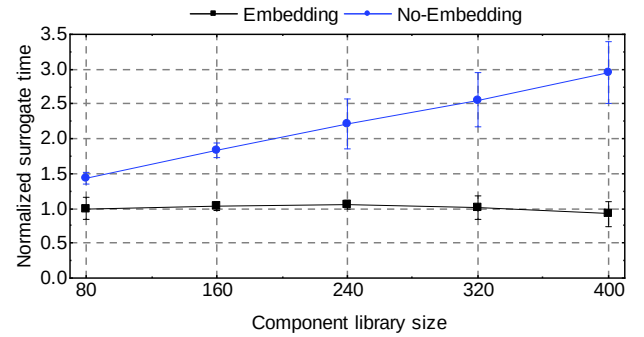


Figure 6: Surrogate model time as the component library grows, with and without the anchor-distance embedding. Times are averaged over the five applications and normalized to the 80-component embedded variant. Markers show the mean and bars show one standard deviation across applications.

Figure 6 compares the surrogate model time for the two representations. The embedded variant remains close to its 80-component baseline, ranging from 0.92 to 1.06 times the baseline as the library grows. In contrast, the non-embedded variant increases from 1.43 to 2.95 times the baseline, nearly doubling as the library grows from 80 to 400 components. Overall, the embedding is already more efficient at our original library size, and its advantage becomes larger as the library grows.

Lower surrogate time is only useful if it does not reduce design quality. Table 5 compares the hypervolume of the embedded and non-embedded representations using their ratio. Except for the 80-component case, the average ratio remains between 0.97 and 1.04, indicating that both representations achieve similar design quality. The higher average ratio of 1.30 at 80 components is mainly due to the activity recognition application, which has a ratio of 1.97, while the other four applications range from 1.01 to 1.35. In summary, the embedding keeps the surrogate dimension fixed as the library grows without noticeably affecting the quality of the resulting designs.

Comp.	Motion	Activity	Mot.+Temp.	Audio	TBI	Avg.
80	1.35	1.97	1.11	1.05	1.01	1.30
160	1.11	0.90	1.10	1.02	1.02	1.03
240	1.07	1.10	1.05	1.00	1.00	1.04
320	0.93	0.91	0.98	1.04	0.98	0.97
400	0.98	0.94	0.94	1.01	1.03	0.98

Table 5: Pareto hypervolume ratio HV_r for the embedded and non-embedded representations as the component library grows. HV_r is the hypervolume obtained with the embedding divided by that obtained without it; values above 1 indicate higher hypervolume with the embedding. Applications are motion streaming (Motion), activity recognition (Activity), motion and temperature monitoring (Mot+Temp), audio event detection (Audio), and traumatic brain injury monitoring (TBI).

G Effect of the Number of Anchor Designs

The number of anchor designs sets the dimension of the embedded space. A larger anchor size gives a finer representation of the design space, but it also gives the Gaussian process more dimensions to fit from the same number of evaluations. We compare 15, 30, and 45 anchor designs on all five applications at a single fidelity (*fid2*) with four seeds each. Because hypervolumes are not comparable across applications, we normalize them within each application across the three anchor sizes, so a value of 1 marks the best of the three and 0 the worst.

Application	Number of anchor designs		
	15	30	45
Motion streaming	0.20±0.26	0.56±0.51	0.74 ±0.43
Activity recognition	0.90 ±0.14	0.00±0.00	0.83±0.31
Motion and temp.	1.00 ±0.00	0.24±0.35	0.23±0.33
Audio event detection	0.46±0.44	0.68 ±0.42	0.40±0.55
Traumatic brain injury	0.56±0.52	0.66 ±0.46	0.55±0.51
Average	0.63 ±0.27	0.43±0.35	0.55±0.43

Table 6: Effect of the number of anchor designs m on Pareto hypervolume (mean $\pm$ std over four seeds). Hypervolumes are normalized within each application across the three anchor sizes, so 0 and 1 denote the worst and best of the three rather than absolute design quality.

As Table 6 shows, no anchor size is best everywhere; 15 and 30 anchors each give the best result on two applications and 45 on one. On average, 15 anchors scores highest at 0.63, and it also has the best worst case at 0.20 against 0.23 for 45 and 0.00 for 30. Larger anchor sets therefore do not give better designs under a fixed evaluation budget, likely because the added dimensions make the surrogate harder to fit. We use 15 anchor designs throughout the paper because it gives the best average result while keeping the embedded space small.

Application	Measured cost	Projected <i>fid3</i>	Saving (%)
Motion streaming	230±39.2	5000	95.4
Activity recognition	224±77.6	5000	95.5
Motion and temp.	383±88.6	5000	92.3
Audio event detection	257±47.6	5000	94.9
Traumatic brain injury	140±54.7	5000	97.2
Average	247±61.6	5000	95.1

Table 7: Measured resource cost of three-fidelity search (*fid1*, *fid2*, *fid3*) compared with the projected cost of an equivalent single-fidelity *fid3* search over the same 50-iteration budget (mean $\pm$ std over four seeds).

H Preference Effect Across the Pareto Front

Figure 7 shows the same effect across the whole front rather than at the rank-1 design alone.

As the preference shifts toward weighted energy, the median weighted energy decreases and its box narrows, while the median latency increases and its box widens. The narrowing indicates that the search is concentrating on a smaller region of the front, which is the intended effect of the preference weights. The trend breaks at preference 3 in (a) and (c), preference 2 in (g) and (j), and preferences 4 and 5 in (k). One reason is that the two objectives are not strictly opposed: a design with low latency does not necessarily have high weighted energy, so a candidate found while favoring one objective can happen to be good for the other objective as well.

I Evaluation Metric Definitions

Cost saved and hypervolume loss. In Section 5.3 we report the resource cost saved and the hypervolume lost by multi-fidelity search relative to the single-fidelity *fid2* run of the same method, computed as

$$\text{Cost saved} = \frac{C_{\text{single-fidelity}} - C_{\text{multi-fidelity}}}{C_{\text{single-fidelity}}} \times 100 \quad (26)$$

$$\text{HV loss} = \frac{HV_{\text{single-fidelity}} - HV_{\text{multi-fidelity}}}{HV_{\text{single-fidelity}}} \times 100 \quad (27)$$

A negative hypervolume loss means the multi-fidelity front is better than the single-fidelity one.

Rank-1 design score. In Section 5.4 we identify the design on the Pareto front that best matches a given application preference by scoring every design with the weighted sum of its normalized objectives and selecting the lowest-scoring one:

$$\text{Score} = p_{E \cdot W} (E \cdot W)_{\text{norm}} + p_L L_{\text{norm}} \quad (28)$$

where

$$(E \cdot W)_{\text{norm}} = \frac{(E \cdot W) - (E \cdot W)_{\min}}{(E \cdot W)_{\max} - (E \cdot W)_{\min}},$$

$$L_{\text{norm}} = \frac{L - L_{\min}}{L_{\max} - L_{\min}}$$

Both objectives are minimized, so the lowest score identifies the design most aligned with the stated preference.

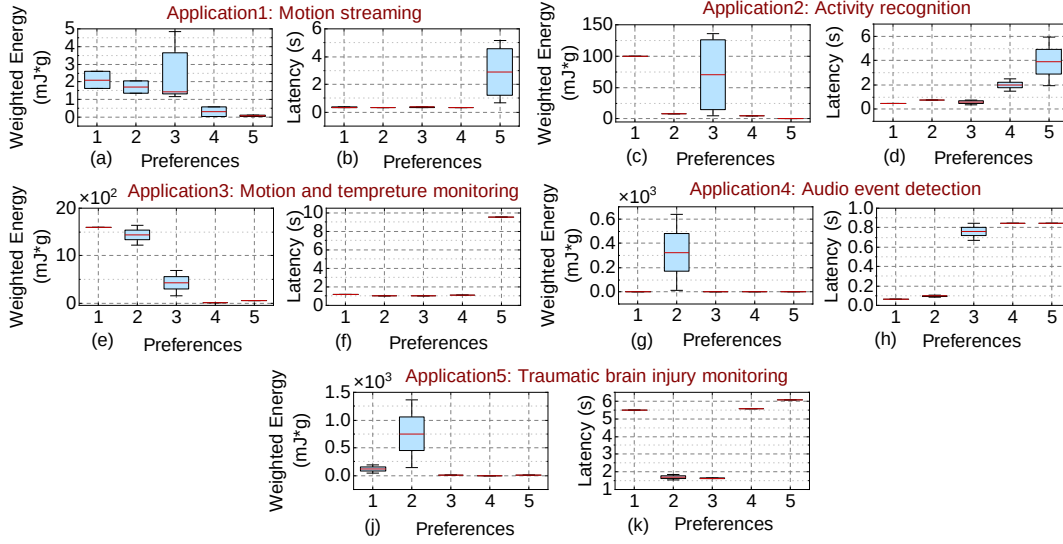


Figure 7: Distribution of weighted energy and latency among the designs on the Pareto front for each application under five application preferences. The preferences $(p_{E.W}, p_L)$ are 1 (0, 1), 2 (0.25, 0.75), 3 (0.5, 0.5), 4 (0.75, 0.25), and 5 (1, 0). Preference 1 optimizes latency alone, and preference 5 optimizes weighted energy alone.

Participant	Overall	Usability	Learnability
P1	22.5	25.00	12.5
P2	72.5	75.00	62.5
P3	75.0	75.00	75.0
P4	70.0	68.75	75.0
P5	75.0	81.25	50.0
P6	55.0	62.50	25.0
Mean	61.7	64.58	50.0
Std. dev.	20.6	20.41	26.2

Table 8: System Usability Scale (SUS) scores from six participants.

Participant	Pragmatic	Hedonic	Overall
P1	-1.25	2.00	0.38
P2	1.75	0.75	1.25
P3	1.00	1.25	1.12
P4	1.50	1.00	1.25
P5	3.00	3.00	3.00
P6	0.75	2.50	1.62
Mean	1.12	1.75	1.44
Std. dev.	1.40	0.89	0.87

Table 9: Short User Experience Questionnaire (UEQ-S) scores from six participants.

J Three-Fidelity Resource Cost

Table 7 reports the measured resource cost of the three-fidelity search for each application, together with the projected cost of an equivalent single-fidelity *fid3* search over the same budget. The measured cost does not depend on the size of the design space. Traumatic brain injury monitoring has the largest design space but the lowest cost, while motion and temperature monitoring has the highest cost. Instead, the cost depends on how often the optimizer selects designs with physically available components, which also explains the relatively large standard deviations.

K Per-Participant Survey Scores

Tables 8 and 9 report the individual scores of the six participants. Each score covers both implemented designs, since the current protocol uses a single combined instrument.

L Component Inventory for Physical Measurement

We currently hold 11 of the 80 components in the library: three processors and two from each of the other categories. These allow 12 to 48 measurable designs per application, which is smaller than the 50-evaluation budget and therefore too small for a single-fidelity *fid3* search. Adding 14 components, so that five are available in each category, raises the measurable space to between 125 and 3,125 designs per application, which is large enough for such a search and also reduces how often a design selected for *fid3* is skipped because a component is unavailable. Table 10 lists the components we currently hold for *fid3* measurement, the inventory planned for the first deployment step, and the number of designs each inventory allows us to measure for every application. Table 11 lists the specific components selected for purchase. They are chosen to widen the range of weight and power covered by the physically available sub-library rather

Category	In library	Available	Planned
Processors	35	3	5
Accelerometers	19	2	5
Communication modules	16	2	5
Temperature sensors	5	2	5
Microphones	5	2	5
Total	80	11	25
<i>Measurable designs</i>			
Motion streaming		12	125
Activity recognition		12	125
Motion and temperature		24	625
Audio event detection		12	125
Traumatic brain injury		48	3,125

Table 10: Component inventory available for *fid3* measurement, the inventory planned to enable a single-fidelity *fid3* search, and the resulting number of measurable designs per application.

Category	Components
Processors	STM32L476, STM32H562
Accelerometers	BMA400, LIS2DW12, ADXL345
Comm. modules	nRF52840, ESP32, HC05
Temp. sensors	BMP280, LPS25HBTR, MAX30205
Microphones	SPH0645LM4H-B, IM69D130, ICS-43432

Table 11: Components selected for purchase. Each is chosen to extend the range of weight and power covered by the physically available sub-library.

than only to increase its size, so that physical measurements reflect more of the trade-offs present in the full library.